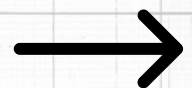
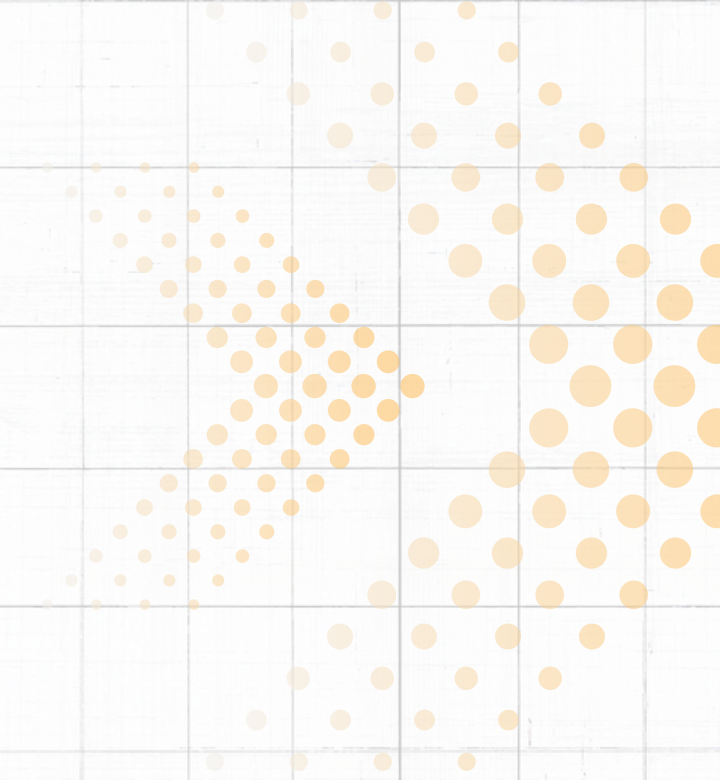
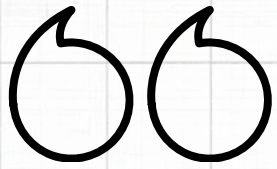


The Universal FSM Design Guide

**4 stage metal
framework**

By : Aadarsh K A S



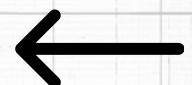


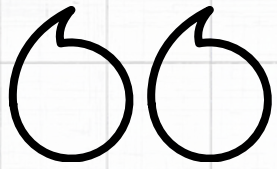
The Black-Box Spec!

Don't think about logic yet. Draw a physical box and list every single Input and Output.

Define your module boundaries before you write a single line of code.

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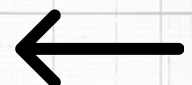


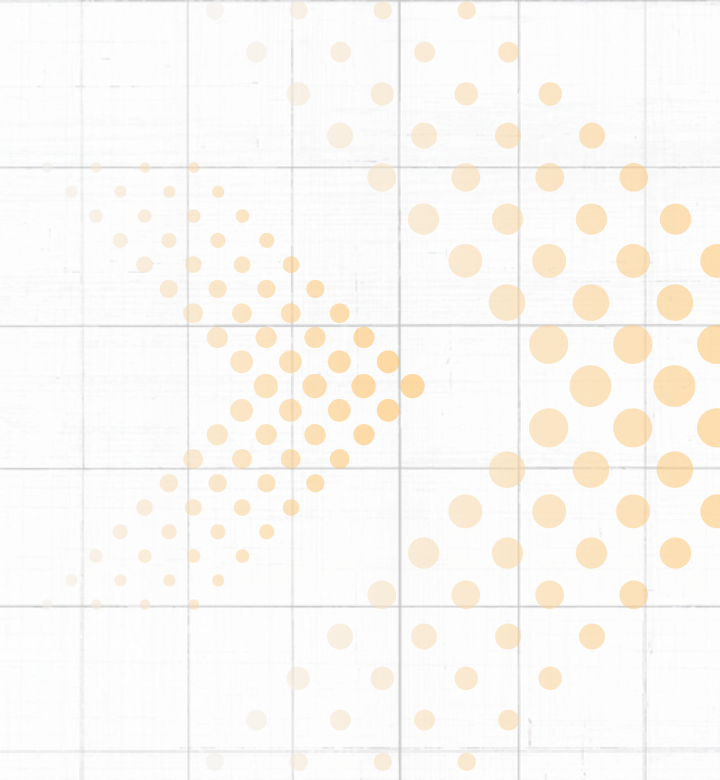
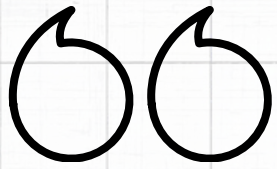
The Visual Map

Draw your State Transition Diagram. Use circles for states and arrows for logic flow.

If the logic doesn't make sense on paper, it won't work in silicon. Sketch it first.

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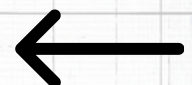


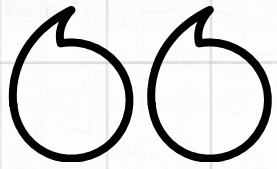


The 3-Block Template

- **Sequential Reg:** State updates.
- **Combo Logic:** Next-state transitions.
- **Output Logic:** Clear, separate signals

Keep memory and logic separate for clean, synthesis-ready code.



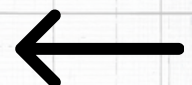


The Final Checklist

Before you simulate, run the "Latch Killer" check:

- Did I use default cases?
- Are my assignments ($\leq$ vs $=$) correct?
- Is every state accounted for?

A 5-minute mental double-check saves 5 hours of debugging.





Example

```
// Block 1: State Register (Sequential)
always @(posedge clk or negedge rstn) begin
    if (!rstn)
        state <= OFF;
    else
        state <= next_state;
end
```

```
// Block 2: Next-State Decoder (Combinational)
always @(*) begin
    next_state = state; // Default: stay put
    case (state)
        OFF: if (button_press) next_state = ON;
        ON:  if (button_press) next_state = OFF;
    endcase
end
```

```
// Block 3: Output Decoder (Combinational)
always @(*) begin
    light_out = (state == ON); // Light is 1 when ON
end
```

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